



Dilshan Sanjaya

BET(Hons) in Instrumentation and Automation Technology

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ABOUT ME

Motivated undergraduate in Instrumentation and Automation Technology with a strong desire to gain hands-on experience and technical expertise. Passionate about applying academic knowledge to real-world application in embedded systems, control systems, and innovative automation solutions. Seeking an opportunity to apply technical knowledge and acquire valuable experience.

EDUCATION

- **Faculty of Technology, University of Colombo** 2022 - 2027 (Expected)
Bachelor of Engineering Technology (Hons) in Instrumentation and Automation Technology
- **H/St. Mary's College**
GCE(A/L) Examination- Engineering Technology (Z-score -2.1275)
Science for Technology – A
Engineering Technology – B
Information and Communication Technology - C

EXPERIENCE

- **Cargills Convenient Foods Ltd** February 2025 - July 2025
Industrial Trainee – Engineering Department
Worked on engineering projects and supported daily maintenance tasks.
- **Greenery Tech (Pvt) Ltd** January 2022 - July 2022
Technical Assistant (Trainee)
Assisted in the maintenance and installation of RO plant machinery, and supported repair and troubleshooting of domestic RO water purification systems.

TECHNICAL SKILLS

- **Electrical & Electronics:** Circuit Design, Hardware Prototyping
- **CAD & 3D Modeling:** AutoCAD, SOLIDWORKS
- **PCB Design & Simulation:** Simul IDE, EasyEDA Pro
- **Microcontroller Programming:** ESP32, Arduino Uno
- **Web & Mobile Development:** React, React Native (Expo), Firebase
- **Familiar With:** PLC Programming, Pneumatic Schematics
- **Office Applications:** Microsoft Word, Excel, PowerPoint

PROJECTS

- **Research Project - Plant Electrophysiological Signal Analysis under Different Salinity Level** November 2025 - June 2026
Designed data acquisition circuit to analyze electrophysiological signals of five tomato cultivars under salinity stress. Developed a feature-extraction machine learning model to identify stress levels and evaluate the most suitable cultivar.
- **PID Monitoring System for Cooling Room** March 2025 - July 2025
Built a real-time PID monitoring system for a cooling room using an ESP32-S3 board to track and control temperature. Created a web-based system to display live sensor data and monitor the cooling process efficiently.
- **Automated Sausage Packet Counting System** March 2025 - June 2025
Built an automated packet counting system using an ESP32-S3, a laser beam, and a barcode reader to accurately track production line output. Created a web-based display system integrated with Firebase to monitor daily counts in real-time and improve packaging efficiency.
- **Dual-Mode Star-Delta Starter (Manual/PLC)** October 2025 - December 2025
Built a three-phase motor starter panel as a group project, featuring both manual relay and automated PLC control modes. Programmed the ladder logic and wired critical safety features, including mechanical interlocks and an emergency stop, to protect the equipment.
- **Data Acquisition System** November 2023 - December 2023
Developed a real-time data acquisition (DAQ) system using an ESP32 to monitor smart greenhouse sensors.

SOFT SKILLS

- Problem-Solving
- Teamwork and Collaboration
- Time Management
- Self-Learning
- Effective Communication

LANGUAGES

- Sinhala
- English

REFERENCES

Dr. B. L. Sanjaya Thilakarathne
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Department of Instrumentation and
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University of Colombo
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